The *what*, *why*, and *how* of open science



SORTEE

Society for Open, Reliable, and Transparent Ecology and Evolutionary Biology



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Who are we?

O'Dea *et al. BMC Biology* (2021) 19:68
https://doi.org/10.1186/s12915-021-01006-3

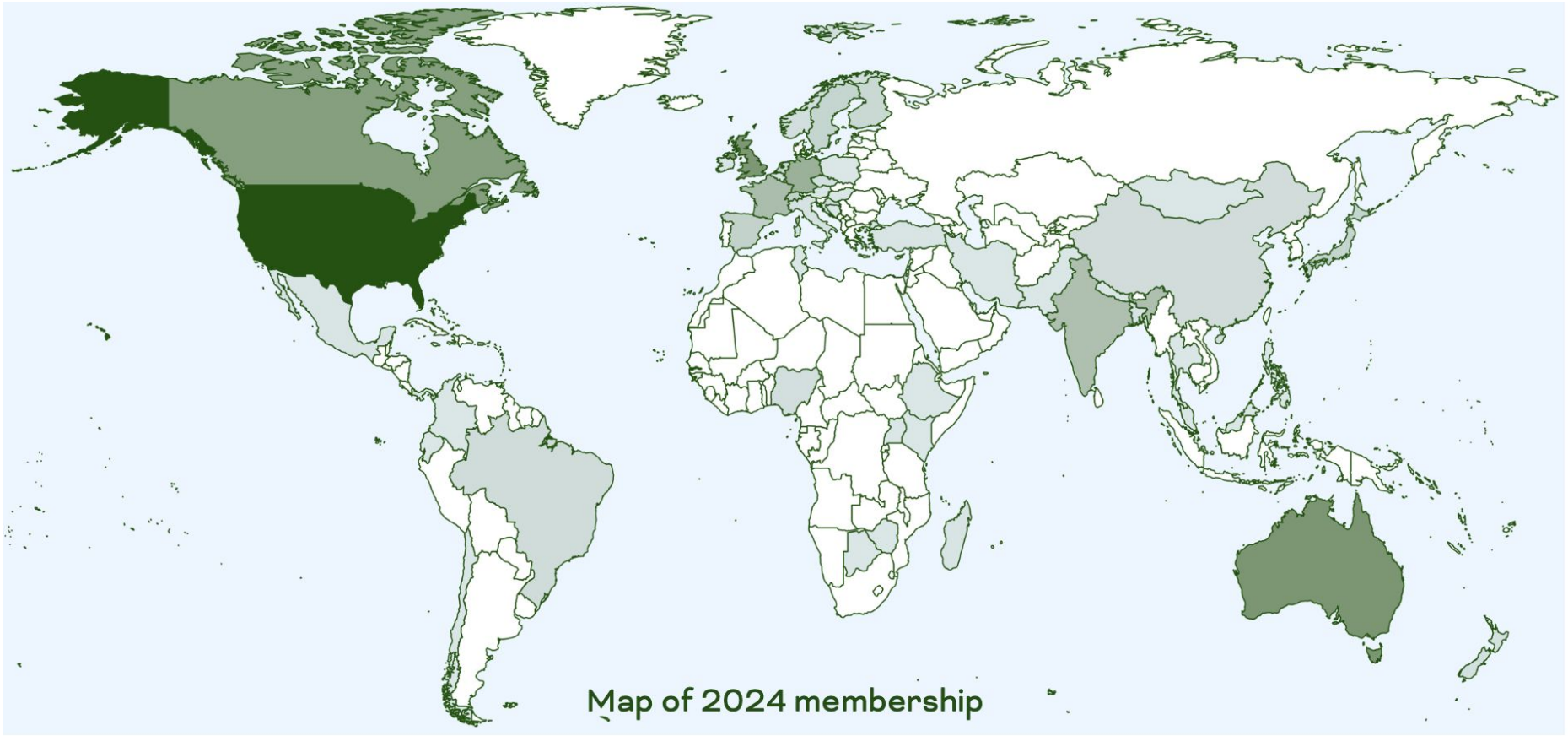
BMC Biology

COMMENT Open Access

Towards open, reliable, and transparent ecology and evolutionary biology

Rose E. O'Dea[1*], Timothy H. Parker[2], Yung En Chee[3], Antica Culina[4], Szymon M. Drobniak[1,5], David H. Duncan[6], Fiona Fidler[6,7], Elliot Gould[6], Malika Ihle[8], Clint D. Kelly[9], Malgorzata Lagisz[1], Dominique G. Roche[10], Alfredo Sánchez-Tójar[11], David P. Wilkinson[6], Bonnie C. Wintle[6] and Shinichi Nakagawa[1]

Who are we?



Who are we?

Open
Code sharing, data sharing, accessibility

Reliable
Code review, reproducibility

Transparent
Publishing reform, pre-registration

Open

Making our research outputs, data, and code available and accessible to all

Data sharing

What?

Findable **A**ccessible **I**nteroperable **R**eusable

Why?

Increase transparency and reproducibility

Enable further research

Avoid research waste

How?

zenodo



DRYAD



figshare

Code sharing

What?

Making the code required for analyses available alongside research outputs

Why?

Increase transparency and reproducibility

Help others learn and improve their own work

Document your research outputs

How?

1 **TADA! Simple guidelines to improve code sharing**

2

3 *Edward R. Ivimey-Cook[1@], Antica Culina[2], Shreya Dimri[3], Matthew J. Grainger[4], Fonti Kar[5,6],*

4 *Malgorzata Lagisz[6,7], Nicholas P. Moran[8], Shinichi Nakagawa[7], Dominique G. Roche[9], Alfredo*

5 *Sánchez-Tójar[3], Saras M. Windecker[10], Joel L. Pick[11]*

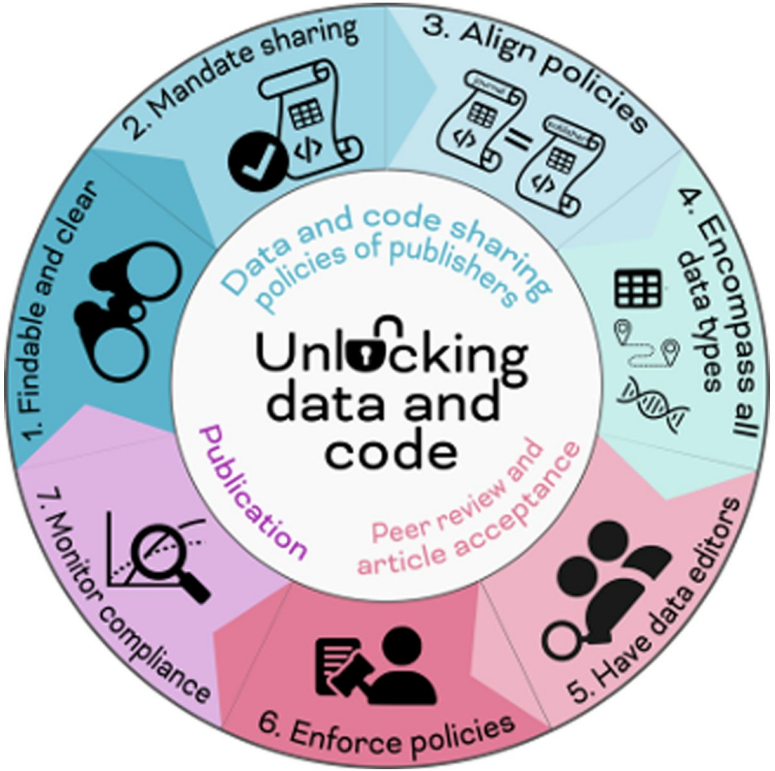


Transferable
Accessible
Documented
Annotated

Publishers

PROCEEDINGS B

From policy to practice: progress towards data- and code-sharing in ecology and evolution



Open science for everyone

What?

Ensuring research is truly open to everyone, no matter who they are or where they work

Why?

Priced out of belonging? Insufficient concessions on membership fees across international societies in ecology and evolution

Malgorzata Lagisz[1,2], Kevin R. Bairos-Novak[3], April Robin Martinig[1,4], Michael G. Bertram[5,6,7], Ayumi Mizuno[1,2], Saeed Shafiei Sabet[8], Matthieu Paquet[9], Manuela S. Santana[10], Eli S. J. Thoré[5,11,12], Nina Trubanová[13], Joanna Rutkowska[14], James A. Orr[3,15], Elina Takola[16], Yefeng Yang[1], Patrice Pottier[1,17], Dylan G. E. Gomes[18,†], Ying-Chi Chan[19], Zhenzhuo Xian[1], Caleb Onoja Akogwu[20,21], Szymon M. Drobniak[1,14] and Shinichi Nakagawa[1,2]

Global North-South science inequalities due to language and funding barriers

Turba, Rachel; Thoré, Eli S.J.; Bertram, Michael G.; Bridg, Hannia; Sabet, Saeed Shafiei; Gamboa, Maribet; Ríos-Orjuela, Juan Camilo; Takola, Elina; Capa Salinas, Jose; Sampaio Franco, Ana Clara; Marín, César

The manifold costs of being a non-native English speaker in science

Tatsuya Amano, Valeria Ramírez-Castañeda, Violeta Berdejo-Espinola, Israel Borokini, Shawan Chowdhury, Marina Golivets, Juan David González-Trujillo, Flavia Montaño-Centellas, Kumar Paudel, Rachel Louise White, Diogo Veríssimo

Reliable

Taking steps to ensure our research is consistent, accurate, and trustworthy

Avoiding questionable research practices (QPRs)

What?

Increasing awareness of practices that can undermine the reliability of our results

Why?

RESEARCH ARTICLE

Questionable research practices in ecology and evolution

Hannah Fraser[1]*, Tim Parker[2], Shinichi Nakagawa[3], Ashley Barnett[1], Fiona Fidler[1,4]

How?

Conservation Letters
Open Access
A Journal of the Society for Conservation Biology

PERSPECTIVE | Open Access

A toolkit for open and pluralistic conservation science

Mark Burgman, Rafael Chiaravalloti, Fiona Fidler, Yizhong Huan, Marissa McBride, Alexandru Marcoci, Juliet Norman, Ans Vercammen, Bonnie Wintle, Yurong Yu

First published: 30 September 2022 | https://doi.org/10.1111/conl.12919 | Citations: 4

Experimental & computational reproducibility

What?

Making sure our work is truly reproducible, now and well into the future

Why?

Computationally reproducing results from meta-analyses in ecology and evolutionary biology using shared code and data

Steven Kambouris[1,2*], David P. Wilkinson[1,2], Eden T. Smith[1,3], Fiona Fidler[1,2,3]



How?

Code sharing and review

Robust research reporting

Pre-registration

Code reliability

What?

Ensuring code is efficient, readable, useable, and reliable

Why?

Ensure robust results

Help future researchers

Improve trust in science

How?



1 The SORTEE Guidelines for Data and Code Quality
2 Control in Ecology and Evolutionary Biology

3 Joel L. Pick[1*], Bethany J. Allen[2], Benedicte Bachelot[3], Kevin R. Bairos-Novak[4], Jack A. Brand[5,6],
4 Barbara Class[7], Tad Dallas[8], Pietro B. D'Amelio[9], Erola Fenollosa[10], Esteban Fernández-
5 Juricic[11], Dylan G. E. Gomes[12], Matthew J. Grainger[13], Thomas Guillemaud[14], Christian John[15],
6 Ruby Krasnow[16], Malgorzata Lagisz[17,18], Sebastian Lequime[19], Daniel S. Maynard[20], Shinichi
7 Nakagawa[18], Rose E. O'Dea[21], Matthieu Paquet[22], Quentin Petitjean[23], Alfredo Sánchez-Tójar[24],
8 Natalie E. van Dis[25,26], Laura A. B. Wilson[17,27], Edward R. Ivimey Cook[28*]

Implementing code review in the scientific workflow: Insights from ecology and evolutionary biology

Edward R. Ivimey-Cook, Joel L. Pick, Kevin R. Bairos-Novak, Antica Culina, Elliot Gould, Matthew Grainger, Benjamin M. Marshall, David Moreau, Matthieu Paquet, Raphaël Royauté, Alfredo Sánchez-Tójar, Inês Silva, Saras M. Windecker

Transparent

Illuminating the practice of science to improve trust and clarity throughout the process

Peer review & publishing reform

What?

Finding new ways to do science with greater transparency

Why?

'It felt very icky': This scientist's name was used to write fake peer reviews

Elsevier retracts dozens of journal articles that were published based on "fictitious" reviews

How big is science's fake-paper problem?

An unpublished analysis suggests that there are hundreds of thousands of bogus 'paper-mill' articles lurking in the literature.

How?



Perspective | Published: 22 May 2018

Empowering peer reviewers with a checklist to improve transparency

Timothy H. Parker, Simon C. Griffith, Judith L. Bronstein, Fiona Fidler, Susan Foster, Hannah Fraser, Wolfgang Forstmeier, Jessica Gurevitch, Julia Koricheva, Ralf Seppelt, Morgan W. Tingley & Shinichi Nakagawa

From policy to practice: progress towards data- and code-sharing in ecology and evolution

Edward R. Ivimey-Cook†, Alfredo Sánchez-Tójar†, Ilias Berberi, Antica Culina, ... See all authors

Preprinting

What?

Publishing papers on open platforms before peer review

Why?

Share findings immediately

Get feedback early

Diversity research outputs

How?





Preregistration

What?

Specifying and registering your research plan *before* experiments begin

Why?

Improve reproducibility

Avoid questionable practices

Improve credibility

How?

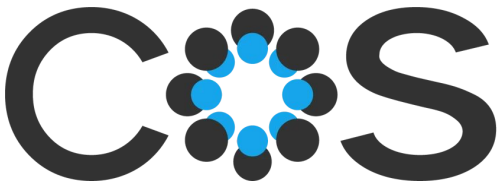
RESEARCH ARTICLE | PSYCHOLOGICAL AND COGNITIVE SCIENCES |

The preregistration revolution

Brian A. Nosek, Charles R. Ebersole, Alexander C. DeHaven, and David T. Mellor Authors Info & Affiliations

Edited by Richard M. Shiffrin, Indiana University, Bloomington, IN, and approved August 28, 2017 (received for review June 15, 2017)

March 12, 2018 | 115 (11) 2600-2606 | https://doi.org/10.1073/pnas.1708274114



Advice for your open science journey

Start small
You don’t need to do everything all at once – small steps make a big difference

Think early
Keeping open science in mind from the start of a project reduces the workload

It doesn’t have to be perfect
Imperfect open science is still more transparent than closed science. Don’t be worried about ‘doing it wrong’

Get involved
Sticking around the community is the best way to learn more

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Interested in getting involved in 2027? Volunteer for a committee!

Workshop & Webinar series: Upcoming talks



Journal of Statistical Ecology

November 18, 8:30 UTC

DAFNEE, a Database of Academia Friendly jourNals in Ecology and Evolution

December 10, 12:00 UTC

Workshop & Webinar series: Youtube

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Questions



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